Gate All Around Field Effect Transistor

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***Abstract*—Recent years have seen the transition of advanced process nodes at leading fabs from FinFET to the gate-all-around FET (GAAFET) architecture, also known as nanosheet and nanoribbons. This comes after several generations of successful scaling with FinFET, which in recent nodes is reaching a fundamental electrostatic limit due to limited gate control of the channel, manifesting as short-channel effects (SCEs) that degrade device performance. This literature review will show that GAAFET improves gate control by surrounding the channel with the gate, combating SCEs and enabling further scaling due to fundamental electrostatic superiority over FinFET. However, the architecture introduces unique problems, from new mobility limiters, to its enormous additional process complexity. We describe the device performance and applications that are enabled by pursuing this technology and why foundries are currently devoting vast resources to its development.**

***Index Terms*—GAAFET, nanosheet, nanoribbon, short-channel effects**

I. Introduction

Moore's law states that the number of transistors on an integrated circuit roughly doubles every two years, leading to an increase in computing power and a decrease in cost over time. To sustain this trend and meet the increasing demands of the electronics industry, there's a need for transistors to shrink further each year. Although Moore's law holds true, shrinking transistors into smaller nodes hasn't been the most straightforward process. The original transistor model, known as the planar model, reached 22nm before its shortcomings, due to short channel effects (SCEs), could no longer be worked around. This led to exploration for new transistor architectures, such as the double-gate MOSFET, delta transistor and FinFET which was also known as the tri-gate, all collectively known as multigate devices. In 1999, Chenming Hu developed the Fin Field Effect Transistor (FinFET). Due to its fin-like channel and three terminal coverage of the channel, the FinFET became a hit in the development of MOSFETs. Its larger surface area of the channel and direct connection to the gate reduced leakage current and gave the gate better control of the electrostatic field. FinFETs drastically improved the size limitations found in the planar transistor from 22nm to 3nm by TSMC as of 2022. Before 2022, the FinFET's physical and performance capabilities stopped at 5nm causing a new kind of architecture to take the top seat.

II. Background

A. FinFET Scaling Limits and Challenges

At smaller FinFET dimensions, physical geometry and short channel effects play a big role in performance loss and unpredictable fet behavior. To keep the gate's electric field uniform during the scaling process, the thickness of the gate oxide must also be reduced. As shown in the equation, $C_{ox} = \frac{\epsilon_{ox} \cdot A}{t_{ox}}$ a thinner oxide increases the capacitance of the gate and ultimately lowers the voltage needed to induce a channel $V = Q/C$. However, scaling the thickness of the oxide below 0.7 nm leads to an increased amount of leakage current due to quantum tunneling [1]. Due to the wave-like nature of electrons, they can pass through a thin oxide when there isn't enough energy for them to go the potential energy barrier (band gap). Below 5nm, the available channel cross-section becomes too small to sustain high drive current [2] and due to patterning limitations and mechanical instability, the fin height cannot be increased indefintely. One of the most critical SCEs is the drain-induced barrier lowering (DIBL). Drain-induced barrier lowering occurs when a high drain to source voltage is applied and causes the drain terminal's depletion region and electric field to enter the channel.This leads to the gate essentially sharing the charge potential of the channel with the drain. As the drain's electric potential (V_{DS}) grows, the depletion region under the terminal widens and its electric field, which was initially confined to the drain's junction, begins to point towards the source junction. As a result of the electric field infiltrating the channel, the potential barrier at the source slopes downward, allowing for electrons to flow into the channel even when the applied gate voltage (V_{GS}) is below the threshold voltage (V_T). Shown in Figure 1, the energy band diagrams for the long channel and short channel devices are drawn for a clearer understanding. When a voltage is applied to the drain terminal, the conduction band drops by the amount of the potential energy of the voltage $E = qV$. In a long channel device, there is a wide gap between the source and drain junctions that allow for any drop in the drain's energy to not affect the source's end of the channel. In a short channel device, the length between the source and drain junctions shorten so that any little changes to one side can easily interfere with the other. In the figure, the short channel device sees a faster drop of the conduction band but because the channel was short to begin with, the drop begins near the

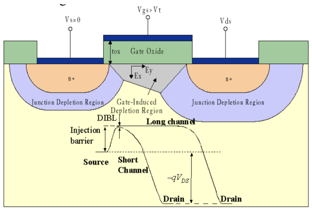


Fig. 1: Illustration of Drain-Induced Barrier Lowering and Energy Band Diagram [4]

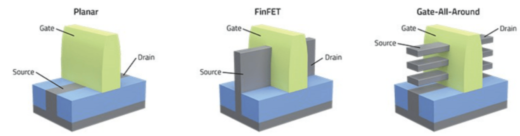


Fig. 2: Architectures of the Planar, FinFET, and Gate-All-Around Nanosheet Transistors

source's junction. This is what is known as DIBL. The energy between the source and channel is called the potential barrier, which is formed by the pn junction between the n-type and p-type semiconductors. A high potential barrier blocks the flow of electrons when $V_{GS} < V_T$. When the drain's conduction band lowers, the electrons will want to move to the lowest available energy state which is towards the drain region, hence the flow of electrons even when the transistor should be off. This causes leakage current and lowers the apparent threshold voltage since there is a flow of current earlier than expected. [3].

Another SCE seen with FinFET scaling is threshold voltage roll off. This effect explains the physical limitations that occur when making the channel length so short that the depletion regions overlap with the channel and encroach one another. When this occurs, the channel is no longer solely controlled by the gate. As both depletion regions enter the channel; the drain, source and gate all share control of the channel potential. Now that the channel potential is influenced by multiple terminals, it needs less gate voltage to invert the channel and turn the transistor on. This again leads to leakage current and unpredictability of the device's switching behavior. A third important SCE is subthreshold slope degradation which is when a MOSFET is unable to suppress channel current in the subthreshold region (V_{GS} <V_T and V_{GS} >0) at at the same rate that the gate voltage decreases. Ideally a MOSFET should have a subthreshold slope of 60 mV/dec at room temperature which means it should drop 10 times for every 60 mV decrease in V_{GS}. In realistic MOSFETs, subthreshold slopes are less steep due to other short channel effects that weaken the gate's control and ultimately blur the lines between the on and off states, resulting in threshold voltages that appear higher. Although the FinFET was a major development for MOSFET technology, the limitations with further scaling has led the industry explore other architectures such as the GAAFET.

B. *Structural Overview of GAAFETs*

The Gate-All-Around Field Effect Transistor (GAAFET) has been a concept researched since the late 1980s [2] and has been commercially used since 2022 with Samsung being the first company to start production. Just as the name suggests, the GAAFET implements a structure where the gate surrounds the transistor's channel on all four of its sides, giving the gate 360 degree control over the channel. The two types of GAAFET channel geometries are the nanosheet and nanowire. The two structures consist of a channel made from silicon, germanium or III-V semiconductors that are completely isolated from the substrate and horizontally stacked and suspended into the gate metal. Around each side of the channel, that is not occupied by the source and drain, is an oxide layer made out of a high-k dielectric that separates the gate metal from the channel. The nanowire channel takes a cylindrical or square-shaped form and its dimensions are relatively small compared to the nanosheet. The nanosheet architecture, which was introduced by IBM's 2nm chip in 2017, is a thin, rectangular plate with a wider dimension than the nanowire. Nanowires were first used because of their electrostatic improvements and while still largely used, nanosheets are becoming the industry standard. Because the nanosheet offers more channel area through its wider dimension, it provides a higher drive current and lower variability [5]. These variabilities come from fabrication imperfections that directly impact the benefits of the GAAFET such as the threshold voltage, drive current, subthreshold swing and more. Further analyses and comparisons are found in [6]. Vertical stacking of channels is an important feature of the GAAFET because each stacked channel acts as an additional conduction path and increases the drive current while preserving the footprint. Nanosheets also offer a tunable width that allows designers to balance performance and power.

C. *Electrostatic Motivation for Gate-All-Around*

The primary purpose of the GAAFET architecture is to achieve a higher electrostatic gate control and drastically suppress the short-channel effects that limit FinFET scaling. By surrounding the channel on all four sides, and its separation from the channel by a high-k dielectric, the gate forms a fully enveloping electric field. This configuration enhances the gate-to-channel capacitive coupling and allows the gate's field to dominate the channel's electrostatics. As a result, the gate can be seen as an electrostatic barrier that shields the impact of the source and drain depletion regions. Although this shielding does not completely eliminate the encroaching of the source/drain fields into the channel, it significantly reduces their impact and mitigates major short-channel effects such as DIBL and threshold voltage roll off. Additionally, the improved gate control directly benefits subthreshold behavior. Because the gate can control the channel potential more efficiently, small changes to the applied gate voltage can greatly influence the onset of conduction and minimize I_{OFF},

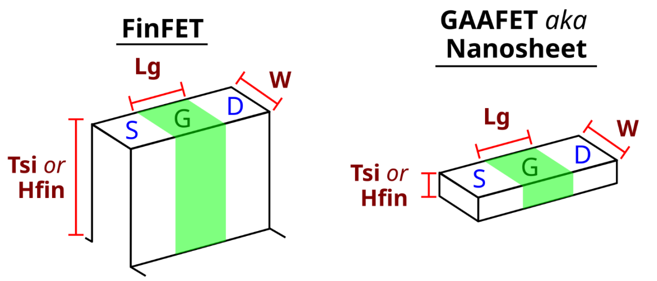


Fig. 3: Definition of equivalent nanosheet and FinFET dimension terms

leading to a steeper subthreshold slope. This improvement enables lower operating voltages and higher power efficiency.

III. GAAFET Physics and Performance

A. Short-Channel Effect Suppression

The scaling advantage of GAAFET devices over equivalent planar and FinFET designs comes from stronger gate to channel coupling by increased gate area vs. silicon volume ratio, and by elimination of the body and associated body effects by isolating the channel from bulk.

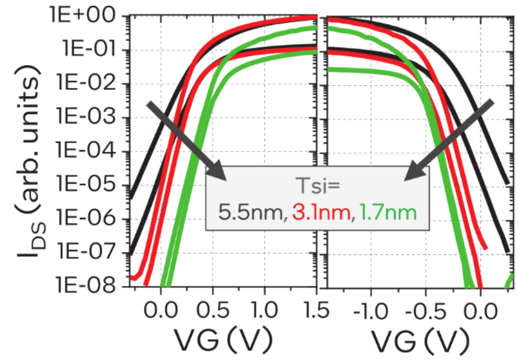
Dimensions that will be used in this discussion of scaling are shown in Fig. 3: T_{Si} or the equivalent FinFET dimension H_{fin} is the height of the channel, L_g is the channel/gate length, and W is the geometric channel width (not to be confused with the electrostatic W_{eff}).

To discuss the device performance benefits of transitioning from FinFET to GAAFET, we will refer the two most prominent geometric differences between the two device types, both of which improve GAAFET performance:

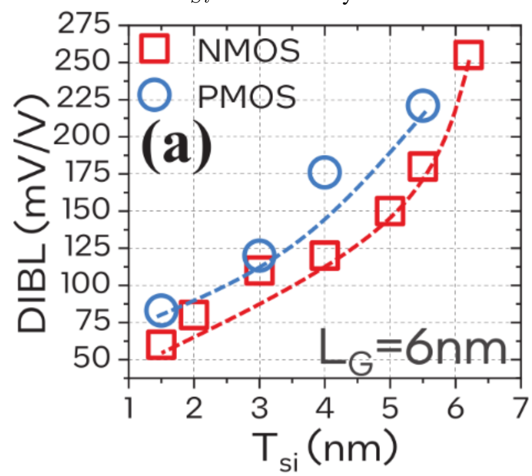
1) Nanosheets and FinFETs have comparable W, but significantly different T_{Si}, or equivalently, "fin height": in FinFET, $T_{Si} \gg W$, while in nanosheets, $T_{Si} < W$.
2) FinFETs are "three-gate-control" devices, approximated as a tall rectangle surrounded on the top and sidewalls by gates. Nanosheets are "four-gate-control" devices, approximated as a squat rectangle surrounded on all walls by gates.

Aside from these differences, the ideal FinFET on SOI and nanosheet designs are identical (processing differences will also be highlighted in Section IV).

Fig. 4 shows short channel effects (SCEs) recover from reducing T_{Si} [7]. In Fig. 4a, $I_d - V_g$ curves were collected from three devices with varying T_{Si}, each measured at $V_{ds} = 50mV, 650mV$, producing two curves per device. This figure shows the predominant performance tradeoff made in GAAFETs: the recovery of DIBL and SS from reducing T_{Si}, at the expense of mobility.



(a) I_d-V_g curves with varying T_{Si}. SS slope recovery and convergence of curves with thinner T_{Si} directly display the recovery of SS and DIBL resp. by the new critical dimension T_{Si} introduced by nanosheets.



(b) DIBL vs. T_{Si}, directly showing recovery of DIBL attributed to unique nanosheet geometry (T_{Si}).

Fig. 4: Empirical relations of SCEs to nanosheet-specific critical dimension T_{Si}. From [7].

- DIBL manifests in $I_d - V_g$ curves as a separation of the subthreshold regions at differing V_d, due to V_d-sensitive subthreshold leakage caused by barrier lowering. As T_{Si} decreases from 5.5nm to 1.7nm, the convergence of the subthreshold curves indicates the improvement in DIBL at smaller T_{Si}.
  This is illustrated more directly by Fig. 4b. DIBL is quantified as the horizontal shift in mV in the subthreshold region per unit V_{ds}, and decreases strongly as T_{Si} decreases;
- The subthreshold slope (SS) steepens when transitioning from the 5.5nm device (black) to the 3.1nm device (red), indicating a suppression of the effect of the S/D depletion region to the channel inversion. The recovery of the slope is due to a recovery of $\frac{\partial \psi_s}{\partial V_g}$ from the fourth gate control that couples to the channel region that is otherwise dominated by the S/D depletion in FinFET. Some recovery of the V_t rolloff is also visible by the shifting of the I_d-V_g knee to larger $|V_g|$;
- At $T_{Si} = 1.7nm$, $I_{ds,max}$ is visibly degraded due to mobility degradation that will be discussed in Section III-C. In particular, for very thin sheets $T_{Si} = 1.7nm$, I_{dsat} is visibly degraded due to a mobility cliff that will be seen

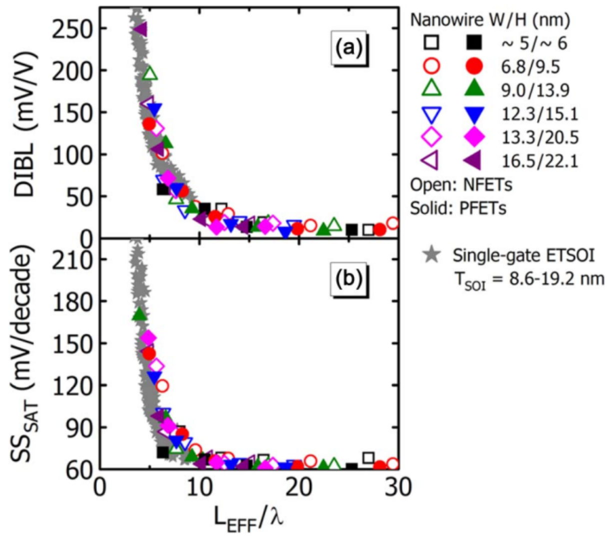


Fig. 5: Universal dependency of SS and DIBL for multiple technologies on the ratio L_{eff}/λ. From [8]

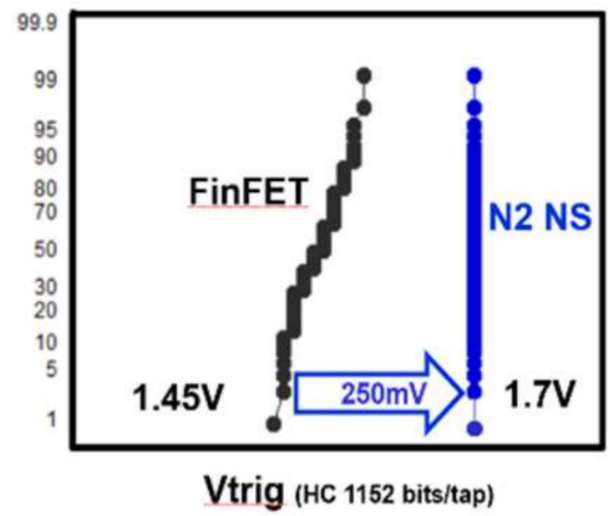


Fig. 6: Latchup/breakdown voltage distributions for SRAM at TSMC's N3 and N2 nodes. From [10].

in Fig. 9a and discussed later.

The physics underlying these improvements are encapsulated in the universal scaling parameter λ, a critical length for electrostatic gate length L_{eff} scaling. This ratio L_{eff}/λ directly dictates SCEs, as shown in Fig. 5, where devices from multiple technologies and process nodes fall on universal curves for both SS and DIBL. Device engineering to combat SCEs is thus dedicated to reducing λ.

In [9], a term λ_x is derived from 3D Poisson solutions for multi-gated FETs (MuGFET), for a given penetration dimension x (e.g. a gate along the height H corresponds to penetration dimension width W):

$$\lambda_x = \sqrt{\frac{\epsilon_{Si}}{2\epsilon_{ox}}\left(1 + \frac{\epsilon_{ox}x}{4\epsilon_{Si}T_{ox}}\right) \cdot x \cdot T_{ox}} \tag{1}$$

Decreasing any dimension x will decreasing λ, leading to the recovery of ideal device characteristics seen in Fig. 4 at small T_{Si}.

The contributions of each dimension is scaled by the number of gates along this dimension. Where a, b denotes the number of gates along the thickness and width of the device, respectively:

$$\lambda_{eff} = \frac{1}{\sqrt{(a/\lambda_{Hfin})^2 + (b/\lambda_{Wfin})^2}} \tag{2}$$

For example, λ for 2-, 3- and 4-gate controls corresponding to double-gate FET (DGFET), FinFET and GAAFET respectively are given by $b = 1$, and $a = 0, 0.5, 1$ respectively. Since increasing a, b gate controls leads to smaller λ_{eff}, GAA offers inherent benefits for SCEs.

Furthermore, GAA recovers W_{eff} by stacking several sheets for an equivalent FinFET. For this same W_{eff}, λ_{eff} is reduced because all smaller dimensions are smaller vs. the equivalent FinFET, and due to the positive relationship between λ and x in Eq. (1).

This is the direct benefit of the GAA architecture, realized by maximizing gate controls and the shrinking of the "fin height" critical dimension H_{fin}.

B. *Reliability Characteristics*

TSMC has assessed its upcoming N2 nanosheet node compared to its previous N3 FinFET on Si node [10] for latchup performance, shown in Fig. 6. Though TSMC uses the term "latchup" loosely, the tightening of the distribution at N2, as well as the otherwise surprising *increase* in V_{trig} for a denser technology, suggests the move to nanosheets has removed the conventional latchup mechanism and is instead limited by another mechanism, possibly avalanche breakdown.

Indeed, because nanosheets are an SOI technology, the parasitic PNPN is eliminated and the technology is latchup-free.

Radiation performance also improves with the isolation from the bulk. Single event transients (SETs) from heavy ion impacts are caused by changes in channel potential due to electron-hole pairs generated along the track of a high-energy, high-Z cosmic ray. The resulting voltage fluctuation can disrupt high speed clocking circuits.

Fig. 7 shows these transients are recovered more quickly in GAAFET than equivalent length FinFET due to smaller ionized volumes compared to ionization of the bulk in FinFET on Si [11].

C. *Mobility Scaling and Scattering Mechanisms*

Preserving high mobility is a pervasive challenge in scaling. In general, high field effects threaten scaling as higher $\mathcal{E}$-fields caused by smaller geometries degrades mobility due to velocity saturation.

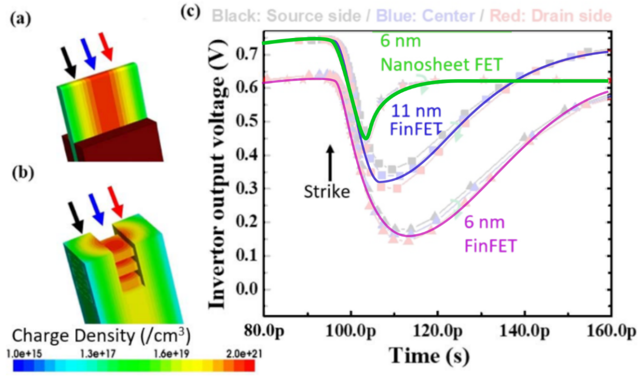


Fig. 7: Simulated SET recovery curves for heavy ion bombardment of GAAFET and two FinFET designs. Adapted from [11]

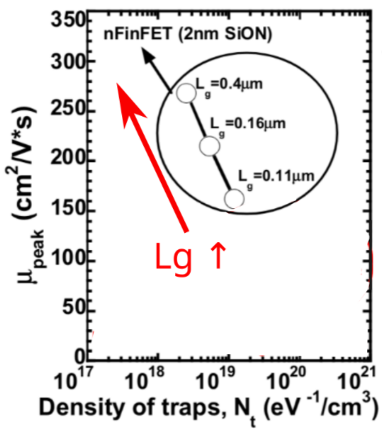
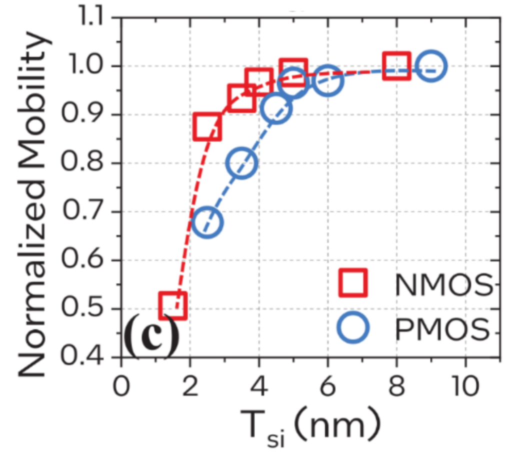
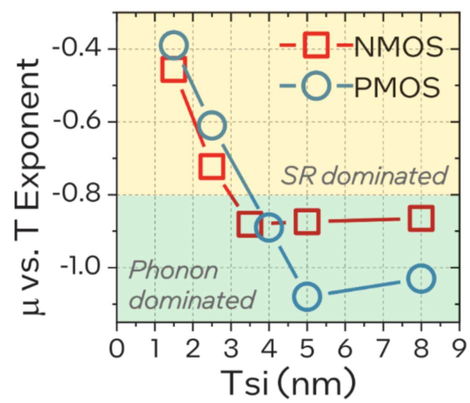


Fig. 8: Mobility as a function of gate length in FinFET, showing increasing gate lengths results in a reduction in N_t interface trap density. From [12]



(a) Mobility as a function of T_{Si} for GAAFET.



(b) Mobility exponential fit parameter to temperature, showing differing mechanisms vs. T_{Si}

Fig. 9: Mobility as a function of T_{Si}. From [7].

One separate effect is identified in extremely short channel devices in [12], indicating fixed interface charges at S/D interfaces of the gate, which do not reduce with scaling, creating increased effective N_t interface trap density and reducing mobility (Fig. 8).

GAAFET scaling in particular has severe mobility constraints induced by tight geometries. While strain and doping engineering has enabled GAAFET technologies to achieve mobilities similar to previous generation FinFETs ($> 200\frac{cm^2}{V \cdot s}$) [7], mobility is constrained by scattering mechanisms introduced in the thin sheet limit that introduces a trade-off of SCE improvements with degraded current drive [7].

In particular, at very small critical dimensions T_{Si}, scattering increases sharply (Fig. 9a) as surface roughness scattering comes to dominate over thermal phonons ($T_{Si} \sim 4nm$) (Fig. 9b). This was identified by measuring mobility as a function of temperature and performing an exponential fit:

- When $T_{Si} > 4nm$, μ is a strong function of temperature, due to strong sensitivity of phonon scattering to the mean free path, which is in turn sensitive to temperature.
- When $T_{Si} < 4nm$, μ becomes less sensitive to temperature which the authors attribute to a dominant and temperature-insensitive surface roughness scattering mechanism, due to the increase in surface area vs. volume for thin sheets.

The relationship between mobility and T_{Si} shown in Fig. 9a is not universal: it is a sensitive function of the processing technology. The main innovation in [7] is pushing the mobility cliff to lower T_{Si} through reduction of the nanoribbon surface roughness compared to prior works, shown in Fig. 10.

IV. Key Process technologies and Design Trade-offs

The fabrication of NS-GAAFET has evolved by building on the conventional HKMG-last FinFET flow. Although approximately 80% [13] of the process remains the same, by adding some key fabrication modules such as Silicon (Si)/Silicon Germanium (SiGe) epitaxy, inner spacer formation, and the channel release process into the current standard processing sequence, undoubtedly introduces new layers of complexities

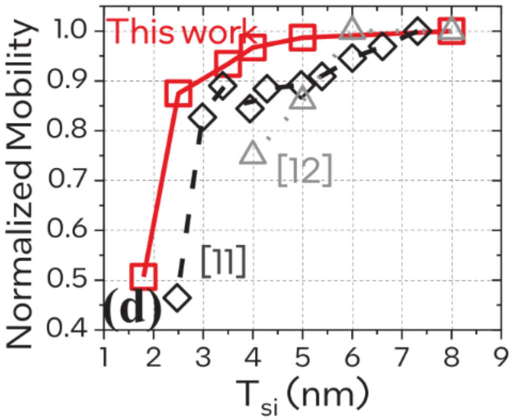


Fig. 10: NMOS mobility as a function of T_{Si}, from [7] compared to previous nanosheet works, showing the improvement of their work and process sensitivity of the relationship.

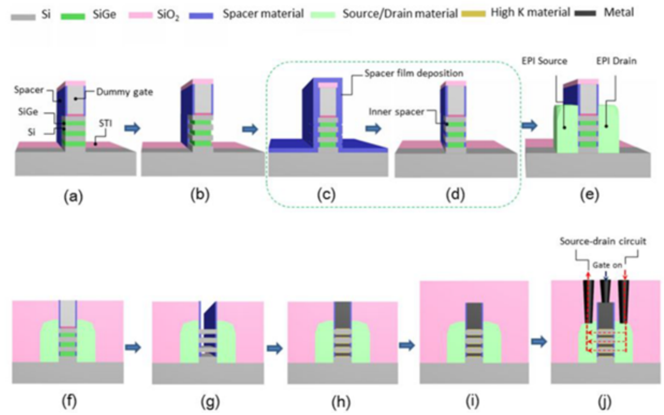


Fig. 11: Process flow of the GAAFET fabrication

and design-trade-offs. These processing steps directly influence the parasitics, sheet integrity, and variability.

A. Key Processing Modules and their Complexities

One of the main process steps that sets GAAFETs apart from other CMOS fabrication is the superlattice epitaxy of the SiGe/Si. The superlattice structure is generally grown on silicon substrates using reduced pressure chemical vapor deposition [13]. To produce the layers with accurate thickness control and well-defined Ge content, the epitaxy is carried out at temperatures below about 650°C, and the process conditions are carefully managed to keep defect levels to a minimum. Maintaining high crystal quality during multilayer epitaxial growth is essential as it directly affects reliability and performance of the device [14].

The inner spacer is another critical component of the GAAFET processing steps, derived from the sidewall engineering principles in FinFETs designed to provide electrical isolation and reduce parasitic channel effects. For the case of stacked SiGe/Si NS devices, the inner spacer specifically acts to minimize the gate to source/drain capacitance (C_{gd}/C_{gs}) [15], [16]. After inner spacer formation, the drain and source epitaxy can be formed on either side of the NS channels, and the replacement metal gate (RMG) formation can follow.

The RMG formation sequence involves several steps, of which the sacrificial channel release, and parts of the high-K metal gate (HKMG) module are specific to the NS devices. The channel release step forms the floating NS structure for the gate material to deposit around the channels. It is done by selectively etching away the sacrificial SiGe layers with advanced reactive ion etching (RIE) with a substantially high selectivity ratio of over $(100:1)$ for $SiGe:Si$. This ensures that the damage done to the Si channels is kept to a minimum [13], [17]. Although the HKMG module is widely used in conventional FinFET fabrication, the geometry of the NS device specifically the spacing between the channels make it challenging to use the current flow due to gate-metal pinch-off concerns [18]. Interface dipole engineering (IDE) offers a volume-less approach to achieve effective work function tuning in place of conventional work function metal (WFM) engineering [13] and is therefore a significant processing step for the NS devices. The key processing steps of the GAAFET fabrication can be seen in Fig. 11.

B. Critical Process Challenges for Nano-Sheet GAAFETs

GAAFET offers superior electrostatic control of the gate as compared with its other CMOS counterparts due to its structural redesign. However, it is important to recognize the challenges that come therein due to this new architecture.

1) Etching process non-idealities: In the superlattice epitaxy of the NS channels, an important parameter to focus on is the percentage of the Ge in the sacrificial Ge layers. By increasing the percentage of the Ge, one can increase the etch rate due to higher structural difference, however, this comes with its own challenges. A higher Ge% results in undesirable Ge diffusion during thermal anneals, and presents epitaxial defects [19]. The Ge% in the sacrificial layers and its subsequent etching rate is shown in Fig.12. It can be seen that lower concentration of Ge results in slower etching rate, but higher Ge concentration results in shallow etch due to diffusion of Ge into channel layers. It is therefore crucial to minimize the Ge% to avoid increasing surface roughness during the etching process, and suffering from surface roughness dominated scattering in thin NS devices as discussed in the earlier sections. The integration of the Source/Drain (SD) contact electrodes with stacked NS channels via complex, simultaneous selective SiGe or Si epitaxy often compromises the conductance pathway due to the misalignment of the NS channels with the electrodes. This leads to high parasitic resistances and degraded driving currents. Landing Pad (LP) technology is a proven method for reducing this parasitic resistance and effectively boosting the device's current performance [20]. Overall, if the etching selectivity for the device processes isn't maintained, channels may be damaged leading to significant **yield** and **reliability** issues.

2) Sub-Fin Leakage: Another aspect in the structure of the GAAFET that introduces challenges is the ”fat-fin” effect

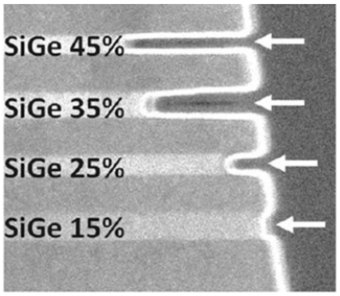


Fig. 12: Etching rate at different Ge%

shown in Fig. 13a that is imposed by the silicon body being too thick to be depleted by the gate, violating the thin body condition:

$$T_{Si} < W_{dep} \tag{3}$$

Where T_{Si} is the thickness of the silicon sheet, and W_{dep} is the depletion width. If the thickness of the fin is larger than the W_{dep}, the gate cannot establish enough depletion charge to fully deplete the fin, therefore leaving it neutral [21]. This incomplete depletion is seen as a saturation of the W_{dep} at a lower value and therefore increases the depletion capacitance C_{dep}. Eqn. 4 defines the SS, where C_{ox} is the oxide capacitance and q is the magnitude of electric charge, both of which remain constant. So from Eqn. 4 it can be seen that the larger C_{dep} directly modulates the value of SS to be larger, worsening the SCEs experienced by the device.

$$SS = \left(\frac{kT}{q}\ln 10\right)\left(1 + \frac{C_{\text{dep}}}{C_{\text{ox}}}\right) \tag{4}$$

Work has been done to mitigate these SCEs experienced due to the sub-fin leakage by either adding a bottom dielectric isolation shown in Fig. 13 to isolate the channels from the effect of the fin-body [22], or by using newer etching technologies to reduce the thickness of the fin [23], and therefore resulting in complete depletion.

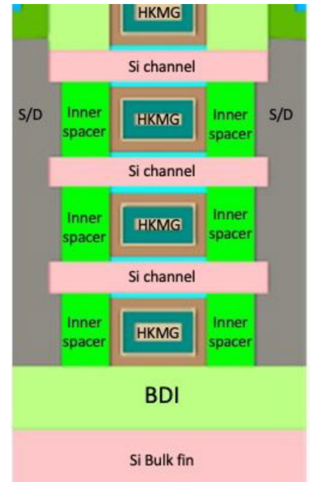
3) Interface Dipole Engineering: Due to the channel spacing being very narrow in NS devices, the conventional WFM process cannot be adopted to tune the threshold voltage V_T and therefore makes the process of the HKMG module significantly more complex. As stated in earlier sections, IDE being a volume-less technique, can be adopted to cleverly adjust the flat-band voltage V_{FB} at the interface and effectively tune the V_T for NS-GAAFET devices [24]. In order to properly use IDE to tune V_T, high temperature annealing is required to drive in the dipoles that are deposited on the high-k layer, which becomes favorable in terms of releasing SD stress, but presents reliability issues [13].

V. Applications and System-Level Impact

A. High-Performance and Mobile SoCs

High performance computing (HPC) and AI accelerator applications require transistors with high switching speeds and power efficiency to meet the demands of the large, complex problems that are solved with GPUs and CPUs.

(a)



(b)

Fig. 13: a) shows the sub-fin effect imposed by thick fin body in NS GAAFETs, b) shows a Bottom Dielectric Isolation implemented for mitigating sub-fin leakage

The GAAFET offers close-to-ideal subthreshold slopes which allow the drain's current to closely follow changes in the gate's voltage, meaning that the device can still operate at a high performance and low leakage current even at low threshold voltages. Because a steeper subthreshold slope provides more current with small changes in V_{GS}, the transistor can charge and discharge processor load capacitors more quickly. This effectively improves switching speeds. The reduction in leakage current and power consumption contribute to the overall improvement of the system's thermal levels in dense computing clusters. In HPC applications, nanosheets are often used to achieve the highest possible clock frequencies while lower operating voltages further reduce the power consumption $P = IV$. While HPC prioritizes speed over power, AI accelerators prioritize energy efficiency for their massive parallel computations to keep energy costs low and heat at a minimal level. Because of the GAA's scalability, designers can integrate more parallel computing units without going over thermal budget. The low operating voltage significantly improves battery life and reduces standby power which also makes it beneficial to smartphones, wearable devices, and IoT systems. The tunable nanosheet width and ability to vertically stack allows designers to precisely choose the MOSFET's behavior when targeting specific power or performance goals. Additionally, GAAFETs are highly researched for their potential in emerging memory technologies such as MRAM and ReRAM, where low leakage is necessary for reducing write energy and improving data retention.

B. 3D Integration and CFET Pathway

As transistor technology begins to reach their physical scaling limits, there has been an increasing need for 3D integration. The transition from a 2D plane to 3D brings an interesting solution to Moore's Law. Rather than scaling a transistor to dimensions that are difficult to achieve, this adaptation provides a higher packing density. Because the GAA nanosheet structure already offers a horizontally stacked channel, the transition to 3D stacking is less complex and is throughly being researched. A promising adaptation is the Complementary Field Effect Transistor (CFET) which vertically stacks an n-type on top of a p-type GAAFET. The CFET is a promising way forward because it takes the electrostatic benefits of the GAAFET and adds it with shorter vertical interconnects. This produces highly power efficient devices that enable faster switching, reduce parasitic capacitances and produce high current, making it a high contender for next generation technology.

VI. Conclusion

GAAFET is showing enormous promise to combat scaling limitations due to short channel effects, while being an enormously complex process technology, that together mark an interesting and challenging stage of device development. Recent process engineering has enabled super-thin sheets that combat SCEs by lowering characteristic penetration depth λ while avoiding compromising mobility due to surface roughness, but the process window is tight, with both effects co-occuring at similar thickness T_{Si}. GAAFET also offers an intrinsic benefit to reliability through isolation from the bulk, which improves latchup and radiation effects. GAAFET processing is very novel, requiring a new channel epitaxy and release step, and by extension, highly selective etches, smooth surface profiles and challenging interstitial oxide and gate metal growth. Issues with this epitaxy and etch directly influence device mobility, impacting typical device performance and yield. Overcoming these process engineering challenges enables a swathe of applications in both high performance and low power, thanks to GAAFET's numerous intrinsic electrostatic and reliability improvements over FinFET. The technology will be interesting to track as it enters mainstream production and scales towards the subnanometer.

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References

[1] L. Qin, C. Li, Y. Wei, Z. Xie, and J. He, "Gate electrostatic controllability enhancement in nanotube gate all around field effect transistor," *AIP Advances*, vol. 13, June 2023.

[2] C. Xie, Y. Zhang, K. Zhang, and L. Z.-R. Guo, "Introduction of gate-all-around fet (gaafet)," *Applied and Computational Engineering*, vol. 28, p. 164–175, Dec. 2023.

[3] J. Qu, H. Zhang, X. Xu, and S. Qin, "Study of drain induced barrier lowering(dibl) effect for strained si nmosfet," *Procedia Engineering*, vol. 16, p. 298–305, 2011.

[4] S. S. Mahato, P. Chakraborty, T. K. Maiti, M. K. Bera, C. Mahata, M. Sengupta, A. Chakraborty, S. K. Sarkar, and C. K. Maiti, "Dibl in short-channel strained-si n-mosfet," in *2008 15th International Symposium on the Physical and Failure Analysis of Integrated Circuits*, pp. 1–4, 2008.

[5] F. I. Sakib, M. A. Hasan, and M. Hossain, "Performance analysis of nanowire and nanosheet ncfets for future technology nodes," *Engineering Research Express*, vol. 3, p. 045044, Dec. 2021.

[6] N. Seoane, J. G. Fernandez, K. Kalna, E. Comesana, and A. Garcia-Loureiro, "Simulations of statistical variability in n-type finfet, nanowire, and nanosheet fets," *IEEE Electron Device Letters*, vol. 42, p. 1416–1419, Oct. 2021.

[7] A. Agrawal, W. Chakraborty, W. Li, H. Ryu, B. Markman, S. H. Hoon, R. K. Paul, C. Y. Huang, S. M. Choi, K. Rho, A. Shu, R. Iglesias, P. Wallace, S. Ghosh, K. L. Cheong, J. L. Hockel, R. Thorman, L. Baumgartel, L. Shoer, V. Mishra, S. Berrada, A. Ashita, C. Weber, B. Obradovic, A. A. Oni, Z. Brooks, N. Franco, J. Kavalieros, and G. Dewey, "Silicon ribbonfet cmos at 6nm gate length," in *2024 IEEE International Electron Devices Meeting (IEDM)*, p. 1–4, IEEE, Dec. 2024.

[8] S. Bangsaruntip, G. M. Cohen, A. Majumdar, and J. W. Sleight, "Universality of short-channel effects in undoped-body silicon nanowire MOSFETs," *IEEE Electron Device Lett.*, vol. 31, pp. 903–905, Sept. 2010.

[9] C.-H. Lin, *Compact Modeling of Nanoscale CMOS*. PhD thesis, University of California at Berkeley., Dec. 2007.

[10] G. Yeap, S. Lin, H. Shang, H. Lin, Y. Peng, M. Wang, P. Wang, C. Lin, K. Yu, W. Lee, H. Chen, D. Lin, B. Yang, C. Yeh, C. Chan, J. Kuo, C.-M. Liu, T. Chiu, M. Wen, T. Lee, C. Chang, R. Chen, P.-H. Huang, C. Hou, Y. Lin, F. Yang, J. Wang, S. Fung, R. Chen, C. Lee, T. Lee, W. Chang, D. Lee, C. Ting, T. Chang, H. Huang, H. Lin, C. Tseng, C. Chang, K. Huang, Y. Lu, C.-H. Chen, C. Chui, K. Chen, M. Tsai, C. Chen, N. Wu, H. Chiang, X. Chen, S. Sun, J. Tzeng, K. Wang, Y. Peng, H. Liao, T. Chen, Y. Cheng, J. Chang, K. Hsieh, A. Cheng, G. Liu, A. Chen, H. Lin, K. Chiang, C. Tsai, H. Wang, W. Sheu, J. Yeh, Y. Chen, C. Lin, J. Wu, M. Cao, L. Juang, F. Lai, Y. Ku, S. Jang, and L. Lu, "2nm platform technology featuring energy-efficient nanosheet transistors and interconnects co-optimized with 3dic for ai, hpc and mobile soc applications," in *2024 IEEE International Electron Devices Meeting (IEDM)*, p. 1–4, IEEE, Dec. 2024.

[11] J. Kim, J.-S. Lee, J.-W. Han, and M. Meyyappan, "Single-event transient in finfets and nanosheet fets," *IEEE Electron Device Letters*, vol. 39, p. 1840–1843, Dec. 2018.

[12] J. Ramos, E. Augendre, A. Kottantharayil, A. Mercha, E. Simoen, M. Rosmeulen, S. Severi, C. Kerner, T. Chiarella, A. Nackaerts, I. Ferain, T. Hoffmann, M. Jurczak, and S. Biesemans, "Experimental evidence of short-channel electron mobility degradation caused by interface charges located at the gate-edge of triple-gate finfets," in *2006 8th International Conference on Solid-State and Integrated Circuit Technology Proceedings*, p. 72–74, IEEE, 2006.

[13] Q. Zhang, Y. Zhang, Y. Luo, and H. Yin, "New structure transistors for advanced technology node cmos ics," *National Science Review*, vol. 11, no. 3, p. nwae008, 2024.

[14] Z. Kong, H. Lin, H. Wang, Y. Song, J. Li, X. Liu, A. Du, Y. Miao, Y. Zhang, Y. Ren, *et al.*, "Multiple sige/si layers epitaxy and sige selective etching for vertically stacked dram," *Journal of Semiconductors*, vol. 44, no. 12, p. 124101, 2023.

[15] N. Loubet, T. Hook, P. Montanini, C.-W. Yeung, S. Kanakasabapathy, M. Guillom, T. Yamashita, J. Zhang, X. Miao, J. Wang, *et al.*, "Stacked nanosheet gate-all-around transistor to enable scaling beyond finfet," in *2017 symposium on VLSI technology*, pp. T230–T231, IEEE, 2017.

[16] H. Mertens, R. Ritzenthaler, V. Pena, G. Santoro, K. Kenis, A. Schulze, E. Litta, S. A. Chew, K. Devriendt, S. Demuynck, *et al.*, "Vertically stacked gate-all-around si nanowire transistors: Key process optimizations and ring oscillator demonstration," in *2017 IEEE international electron devices meeting (IEDM)*, pp. 37–4, IEEE, 2017.

[17] J. Li, Y. Li, N. Zhou, W. Xiong, G. Wang, Q. Zhang, A. Du, J. Gao, Z. Kong, H. Lin, *et al.*, "Study of silicon nitride inner spacer formation in process of gate-all-around nano-transistors," *Nanomaterials*, vol. 10, no. 4, p. 793, 2020.

[18] J. Yao, Y. Wei, S. Yang, H. Yang, G. Xu, Y. Zhang, L. Cao, X. Zhang, Q. Liu, Z. Wu, H. Yin, Q. Zhang, J. Li, and J. Luo, "Record 7(n)+7(p) multiple vts demonstration on gaa si nanosheet n/pfets using wfm-less direct interfacial la/al-dipole technique," in *2022 International Electron Devices Meeting (IEDM)*, pp. 34.2.1–34.2.4, IEEE, Dec. 2022.

[19] N. Loubet, S. Kal, C. Alix, S. Pancharatnam, H. Zhou, C. Durfee, M. Belyansky, N. Haller, K. Watanabe, T. Devarajan, *et al.*, "A novel dry selective etch of sige for the enablement of high performance logic

stacked gate-all-around nanosheet devices," in *2019 IEEE International Electron Devices Meeting (IEDM)*, pp. 11–4, IEEE, 2019.

[20] J. Tian, Y. He, Q. Zhang, C. Wu, L. Cao, J. Yao, S. Mao, Y. Luo, Z. Zhang, Y. Li, *et al.*, "Improving driving current with high-efficiency landing pads technique for reduced parasitic resistance in gate-all-around si nanosheet devices," *ECS Journal of Solid State Science and Technology*, vol. 11, no. 3, p. 035010, 2022.

[21] Y. Taur and T. H. Ning, *Fundamentals of modern VLSI devices*. Cambridge university press, 2021.

[22] S. Mukesh and J. Zhang, "A review of the gate-all-around nanosheet fet process opportunities," *Electronics*, vol. 11, no. 21, p. 3589, 2022.

[23] J. Gu, Q. Zhang, Z. Wu, Y. Luo, L. Cao, Y. Cai, J. Yao, Z. Zhang, G. Xu, H. Yin, *et al.*, "Narrow sub-fin technique for suppressing parasitic-channel effect in stacked nanosheet transistors," *IEEE Journal of the Electron Devices Society*, vol. 10, pp. 35–39, 2021.

[24] D. Zheng, W. Chung, Z. Chen, M. Si, C. Wilk, and P. D. Ye, "Controlling threshold voltage of cmos soi nanowire fets with sub-1 nm dipole layers formed by atomic layer deposition," *IEEE Transactions on Electron Devices*, vol. 69, no. 2, pp. 851–856, 2021.